

WASP-12b

R-2128

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_150116 · created 2026-08-11T23:09:55+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457038.88889 +/- 0.00013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.045
Transit depth (Rp/Rs)^2	2.38 +/- 1.39 [%]
Orbital Inclination (inc)	85.15 +/- 2.84
Airmass coefficient 1 (a1)	0.68 +/- 0.44
Airmass coefficient 2 (a2)	0.09 +/- 0.29
Scatter in the residuals of the lightcurve fit is	66.25 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	3.0
Transit Duration (day)	0.1312 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.045 vs 0.1178 (deviation 0.64σ)
Duration fit vs published	0.1312 d vs 0.125 d (deviation 0.7σ)
Mid-time O–C	0.1 min
Depth SNR	2.7
Residual scatter	66.25 %

Light curve

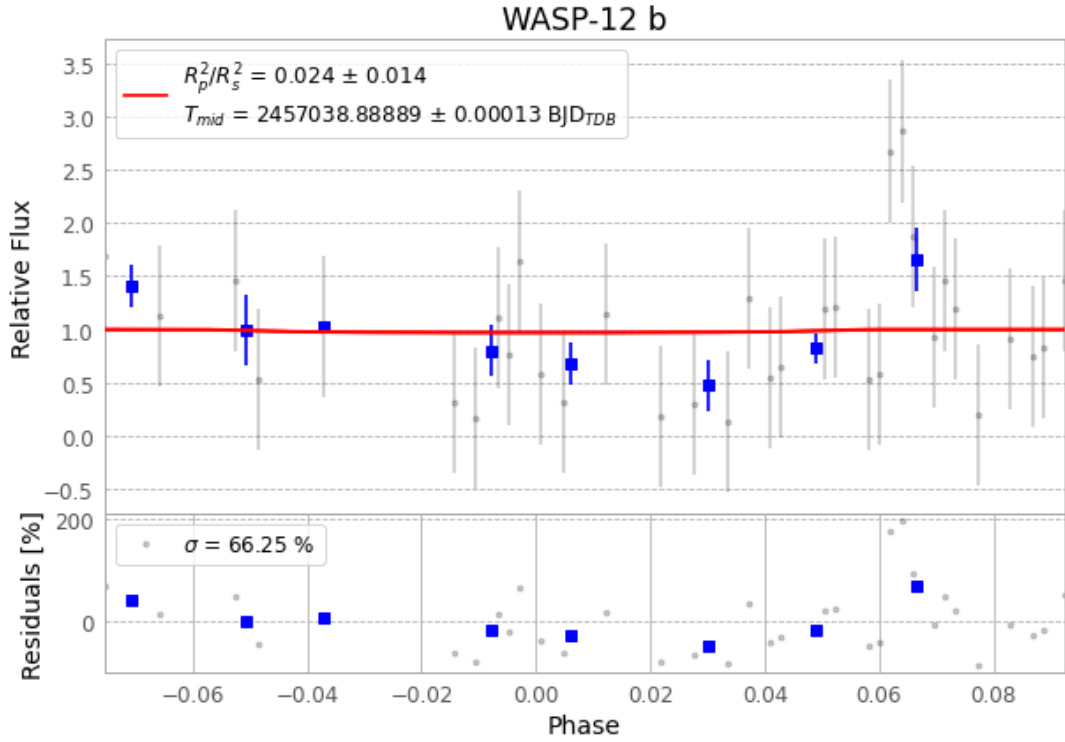


Figure 1 — FinalLightCurve_WASP-12 b_2015-01-16.png (SHA-256 202a5bebf01ca410...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [371, 355], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0973963f490b3cb7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

97 FITS frames (64 MB), WASP-12150116070012.FITS → WASP-12150116114812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-150116.log (27b3e2523d45...), FinalLightCurve_WASP-12 b_2015-01-16.png (202a5bebf01c...), FinalLightCurve_WASP-12 b_2015-01-16.csv (5b6ed3750331...)

Compute resources

Wall time 162.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 23:09Z.